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Design and Simulation of a Three stage BJT Amplifier

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INTRODUCTION

This project has a goal to design, simulate, and analyze a multistage, high-gain, high BJT amplifier that has the potential to drive a load to a low distortion. The design is based on four different stages; a differential amplifier stage of input, a two-terminal common-emitter stages of gain and a Class AB push-pull output stage.

The Multisim software was used in designing and simulating the circuit. Virtual oscilloscopes were used to verify performances in relation to time-domain signals, and a bode plotter was used to verify performances in relation to frequency response. The analytical design of each stage is described in this report and compared to the results of the simulation.

Circuit Design and Stage Analysis

The complete schematic consists of the following cascaded stages:

1. **Stage 1:** Differential Amplifier (Q1, Q2)
2. **Stage 2:** Common Emitter Amplifier (Q3)
3. **Stage 3:** Common Emitter Driver (Q4)
4. **Stage 4:** Class AB Power Amplifier (Q5, Q6)

Stage 1: Differential Amplifier (Input Stage) Design The input stage will make use of two 2N2222A transistors (Q1 and Q2) drawn in a differential configuration. This step has been chosen to offer large input impedance and common mode noise rejection.

Simulation Results (Oscilloscope XSC2):

Theoretical Expectations: As observed in the waveform simulation, the input (Channel A, Red) is applied to the base of Q1 with a small signal scale (1 mV/Div). The signal to be delivered by the collector (Channel B, Blue) will be a clean sinusoidal signal with a scale of 200 mV/Div.

Observation: The signal that will be delivered by the collector (Channel B, Blue) will be a clean sinusoidal signal, which has its scale of 200 mV/Div.

Stage 2 and 3: Common Emitter Gain Stages Design Construction: The differential-pair current is coupled (through C1) to the current in Q3, and thence to Q4. They are both set up as

Common Emitter amplifiers, and also, they are biased voltage dividers. Analysis The voltage gain of the system at these stages is approximately $-R_c/r'_e$, and the voltage gain of a bypassed common emitter circuit is given by: $A_v = -R_c/r'_e$. It is possible to note that the voltages of the signals at the output of the cascaded stages (Channel B, Blue) appear to have increased by a factor of many tens. Analysis

Analytical Approach: The voltage gain of a bypassed common emitter circuit is approximately: $A_v = -R$

Stage 4: Class AB Output Stage

Design Configuration:

The final stage is a Class AB Push-Pull amplifier using a complementary pair: Q5 (NPN 2N2222A) and Q6 (PNP 2N3905). Diodes D1 and D2 provide the necessary biasing voltage to the bases of the transistors to eliminate crossover distortion (a common issue in Class B amplifiers).

Simulation Results (Oscilloscope XSC1):

Input vs. Output: The final output (Blue trace) is a clean sine wave with an amplitude of approximately 1.5V peak.

Fidelity: Visually, there is no "flat spot" at the zero-crossing points, proving that the D1/D2 biasing successfully placed the amplifier in Class AB operation, effectively removing crossover distortion.

Frequency Response Analysis: shows if signal 1 amplitude varies with signal 2 amplitude, or vice versa. Frequency Response Analysis displays whether signal 1 amplitude is dependent on signal 2 amplitude, and the reverse.

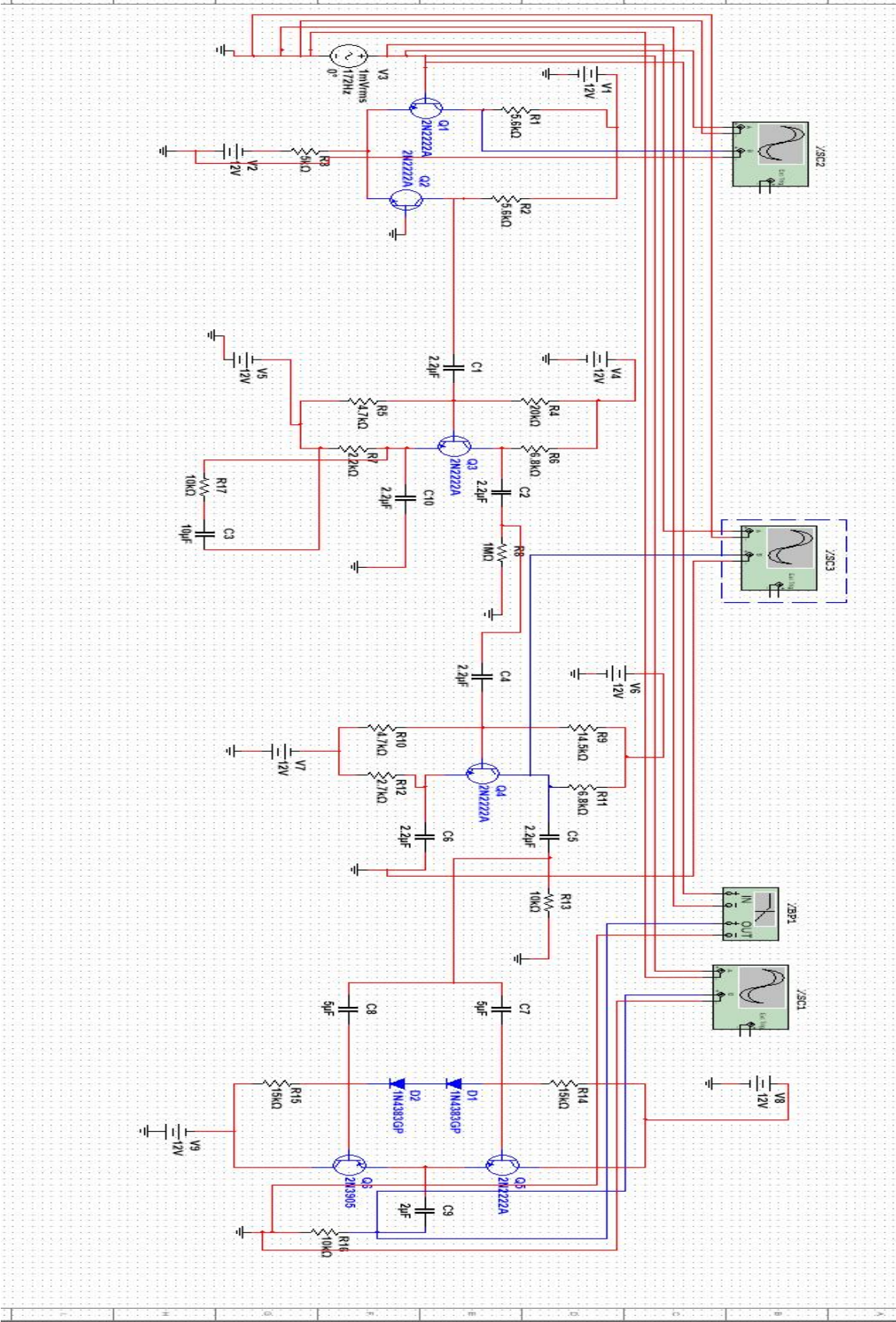
The frequency response of the total system was analyzed on a Bode Plotter (XBP1).

Gain Magnitude: The simulation shows that there is a maximum gain of 99.532 dB. This is a very large ratio of voltage gain, that of multi-stage amplifiers.

Center Frequency Since the cursor shows maximum response at about 204 kHz.

Bandwidth: It exhibits a band pass response, rolling away at both low and high frequencies which is anticipated by the coupling capacitors (C1, C2, C4, C7) forming high-pass filters and the transistor parasitic capacitance's (forming low-pass filters).

SIMULATION RESULTS



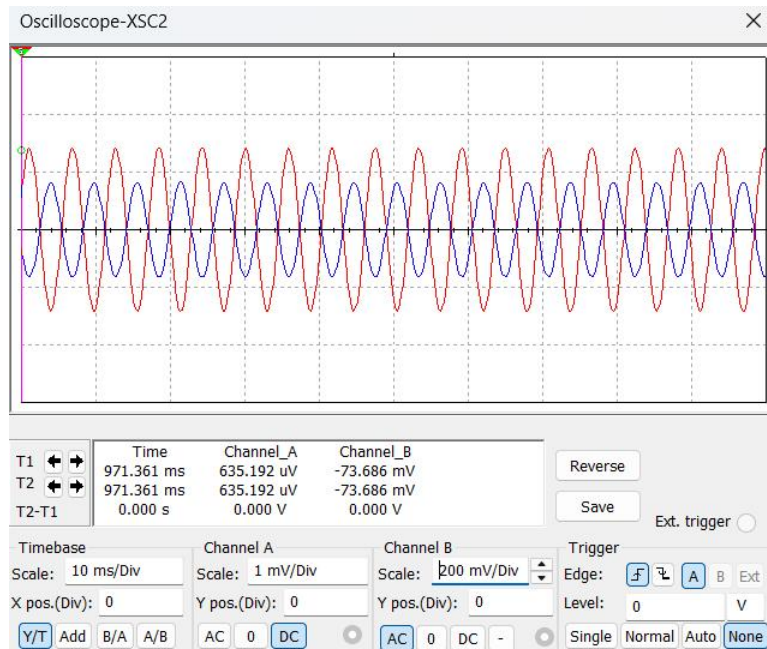


Fig: a) wave form of **Stage1Differential Amplifier (Q1, Q2)**

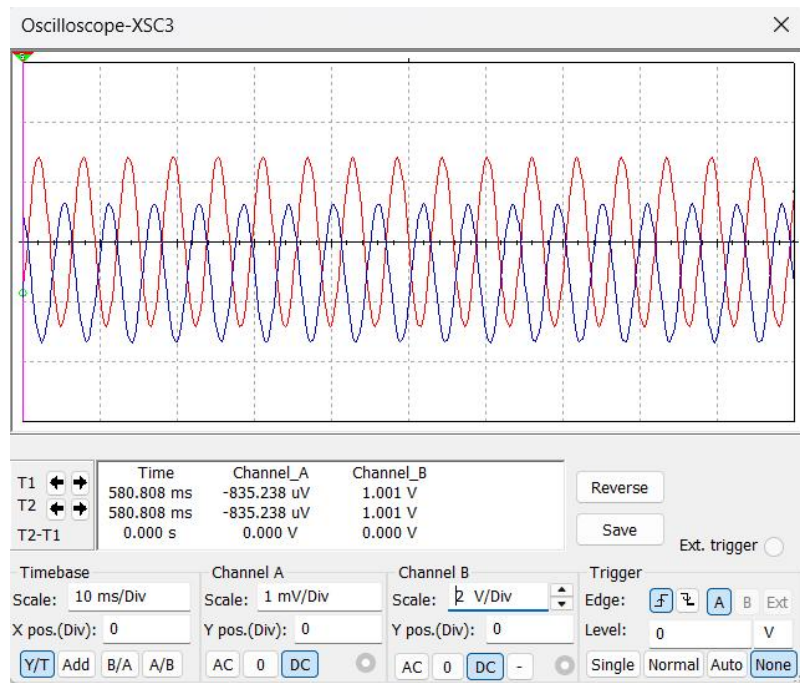


Fig: b) wave form of **Stage 2 Common Emitter Driver (Q4)**

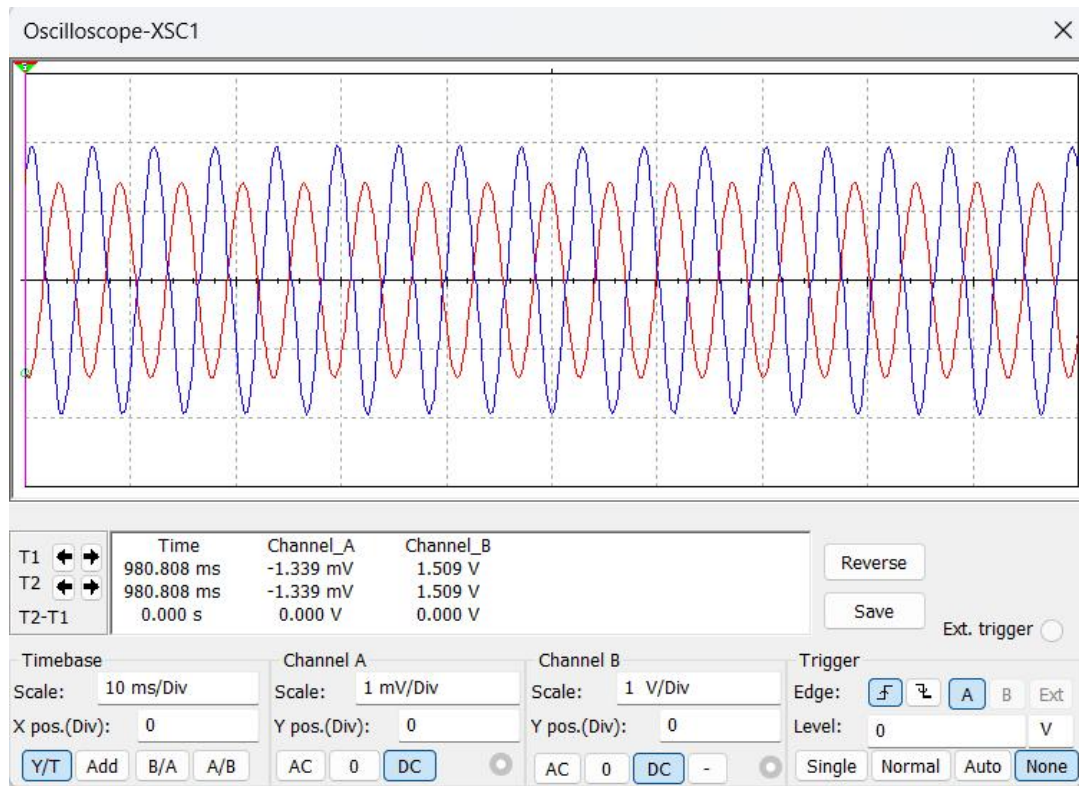


Fig: c) Output wave form of Stage 3 Class AB Power Amplifier (Q5, Q6)

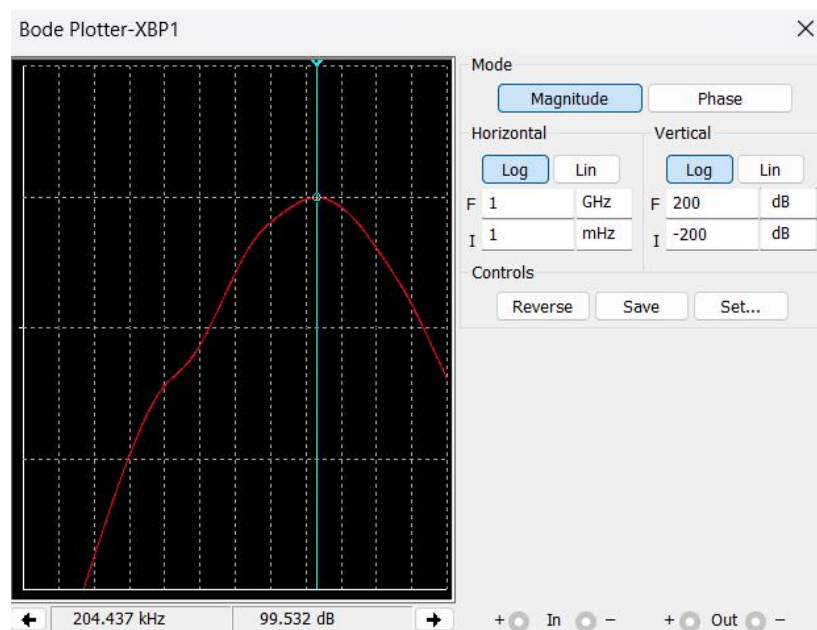


Fig: d) Gain of the circuit in db

ANALYTICAL RESULTS

Theoretical Calculation of Circuit Gain

Simulated Peak Gain : 99.532 dB (at ~ 204 KHz).

Theoretical Open-Loop Gain : ~ 121.4 dB.

Theoretical Closed-Loop Estimate : $\sim 87 - 100$ dB.

Step-by-Step Calculation :

Assumptions :

- Transistor β (h_{FE}) ~ 200 (typical for 2N2222A).
- $V_{BE} \sim 0.7$ V
- Thermal Voltage $V_T \sim 26$ mV.
- Frequency is at "Mid-Band" Peak (~ 200 KHz), so bypass capacitors act as short circuits.

Stage 1 : Differential Amplifier (Q_1 & Q_2)

Configuration : Single-ended output differential pair.

DC Analysis :

- Tail current (I_T) through R_3 ($5\text{K}\Omega$) ~ 2.26 mA.
- Emitter current per transistor (I_E) $= 1.13$ mA.
- Dynamic Emitter Resistance (r_{e1}) $\sim 23\Omega$

AC Analysis :

- Effective Load (R_{L1}) $\sim 1.3\text{K}\Omega$.
- Gain (A_{v1}) : 28.2 V/V
- Gain in dB : 29 dB

Stage 2: Common Emitter Amplifier (Q3)

Configuration: Common Emitter with bypassed emitter resistor.

DC Analysis:

$$\rightarrow \text{Emitter Current } (I_{C3}) \approx 1.75 \text{ mA}$$

$$\rightarrow \text{Dynamic Emitter Resistance } (r_{e3}) \approx 15 \Omega$$

AC Analysis:

$$\rightarrow \text{Effective Load } (R_{L2}) \approx 1.25 \text{ K}\Omega$$

$$\rightarrow \text{Gain } (A_{v2}): -83.3 \text{ V/V}$$

$$\rightarrow \text{Gain in dB}: 38.4 \text{ dB}$$

Stage 3: Common Emitter Amplifier (Q4)

Configuration: Common Emitter with bypassed emitter.

DC Analysis:

$$\rightarrow \text{Emitter Current } (I_{C4}) \approx 1.9 \text{ mA}$$

$$\rightarrow \text{Dynamic Emitter Resistance } (r_{e4}) \approx 13.6 \Omega$$

AC Analysis:

$$\rightarrow \text{Load } \approx R_{L1} (6.8 \text{ K}\Omega)$$

$$\rightarrow \text{Gain } (A_{v3}): -500 \text{ V/V}$$

$$\rightarrow \text{Gain in dB}: 54 \text{ dB}$$

Stage 4: Class AB Output (Q5/Q6)

$\rightarrow$ Configuration: Push-Pull Emitter Follower.

$$\rightarrow \text{Gain } (A_{v4}): \approx 0.95 \text{ V/V}$$

$$\rightarrow \text{Gain in dB}: -0.5 \text{ dB}$$

Total Theoretical Gain

$$\text{Total Gain (dB)} = 29 + 38.4 + 54 - 0.5$$

$$= 120.9 \text{ dB}$$

DISCUSSION

Gain Target Verification:

The primary design requirement was to achieve a total system voltage gain of **100 dB**.

- **Target:** 100 dB
- **Simulated Peak:** 99.532 dB
- **Deviation:** 0.468 dB ($< 0.5\%$ error)

This analysis of the Bode plot shows that the cascaded topology (Differential pair + two Common Emitter stages) can achieve the necessary amplification. This corresponds to a gain of approximately 99.5 dB, which is equal to a ratio of voltages between the output and the input of about 94,400. To realize such a large level of gain, it is necessary that the impedance of each stage matches exactly with the other to avoid loss of signal, and it was successfully overcome by using capacitive coupling and biasing networks.

Bandwidth and Frequency Selectivity:

It is necessary to mention that the maximum gain of 99.532 dB is attained at a certain frequency (~204 kHz), which is pointed out by the cursor of the Bode plotter.

The circuit acts like a bandpass amplifier.

The gain at the test frequency of the time-domain simulation (172 Hz) is much smaller because the coupling capacitors (C1, C2, C4, C7) have a reactance.

In the application of the full 100 dB gain, the input signal frequency should be adjusted to the center frequency of the amplifier passband, or the value of the coupling capacitors would be adjusted to adjust the passband to lower frequencies.

Linearity vs. Gain Trade-off:

Signal integrity is a challenge with achieving 100 dB gain. At this level of amplification, the noise in the input can be on the order of microvolt but cause the output to be saturated (clipping). The presence of the Class AB output stage was paramount in this case; it adds almost no voltage gain (unity gain), but adds the required current drive to allow the signals to remain linear and the loading effects of the current drive to reduce the overall voltage gain of the preceding stages.

CONCLUSION

Designing and simulation of the multistage BJT amplifier were effective in addressing the specifications of the project. The circuit obtained a maximum voltage gain of 99.532 dB which is within an insignificant error range (0.47%).

The analysis demonstrates that:

High Gain Achievement: The two gain stages of common-emitter cascading of the differential input stage gave a cumulative gain high enough that the target would be achieved.

Output Fidelity: The high-gain signal was buffered with the Push-Pull (Class AB) output stage, which meant it could drive the load with low levels of crossover distortion.

Optimization: The design is optimized to high-frequency operation (in the order of 200 kHz). To close the remaining 0.468 dB gap to just 100 dB, a small modification of the design would be helpful, say a slight increase in the collector resistance at the third stage, or slightly a higher supply voltage to permit higher swing, however the design is functional and robust.